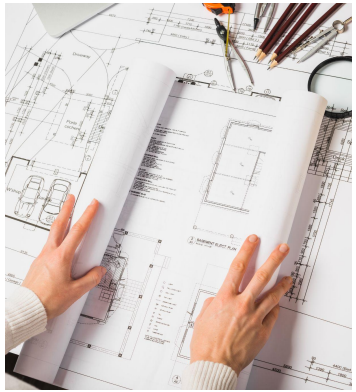
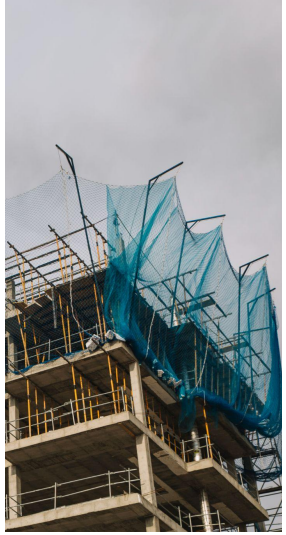
Agile Software Engineering

Software Engineering Course – 2
2024-2025











Agenda

01

Requirements

02

User Personas

03

Features

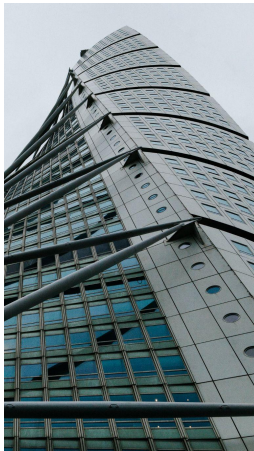


04

Product Backlog

05

Managing technical risk



Requirements Gathering

Informal Requirements Gathering	Formal Requirements Gathering
Personal Experience	Survey
Empirical Evicence	Market Research
Business Analysis	Business Analysis

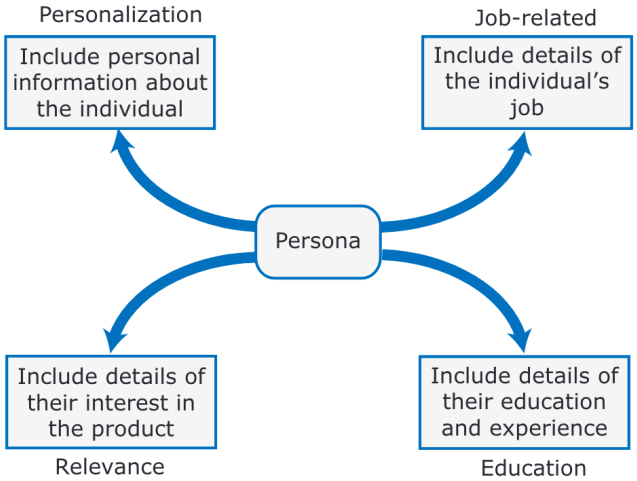


FACEMASH

User Personas

Who are we building this for?

Figure 3.4 Persona descriptions



User Personas

Who are we building this for?

Figure 3.4 Persona descriptions

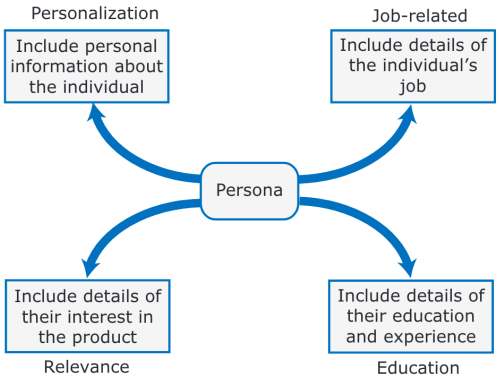


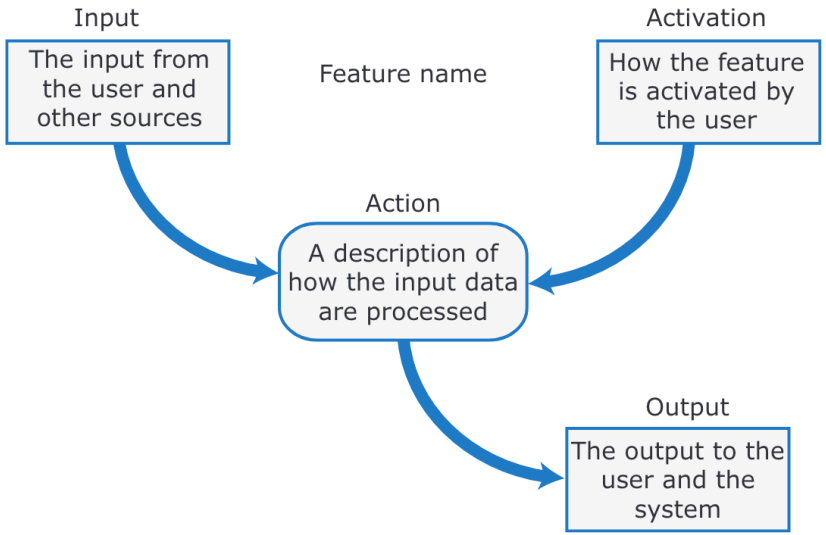
Table 3.4 A persona for an IT technician

Elena, a school IT technician
Elena, age 28, is a senior IT technician in a large secondary school (high school) in Glasgow with over 2000 students. Originally from Poland, she has a diploma in electronics from Potsdam University. She moved to Scotland in 2011 after being unemployed for a year after graduation. She has a Scottish partner, no children, and hopes to develop her career in Scotland. She was originally appointed as a junior technician but was promoted, in 2014, to a senior post responsible for all the school computers. Although not involved directly in teaching, Elena is often called on to help in computer science classes. She is a competent Python programmer and is a "power user" of digital technologies. She has a long-term career goal of becoming a technical expert in digital learning technologies and being involved in their development. She wants to become an expert in the iLearn system and sees it as an experimental platform for supporting new uses for digital learning.

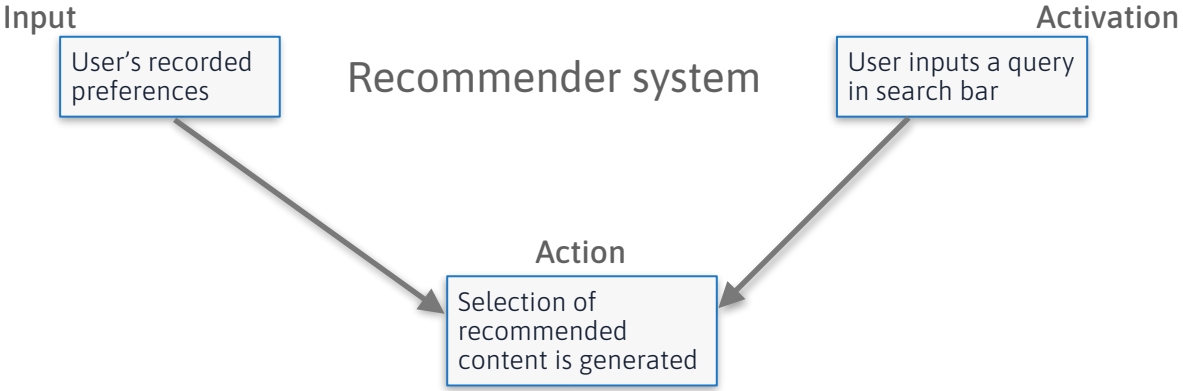
Derive scenarios from User Personas (ESP p.69 - 75)

Thinking in Features

Figure 3.2 Feature description

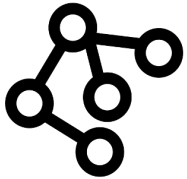


Thinking in Features




```
function ACTIVATION (string INPUT){
    //...
    //ACTION
    //...
    return OUTPUT;
}
```

Anatomy of a User Story



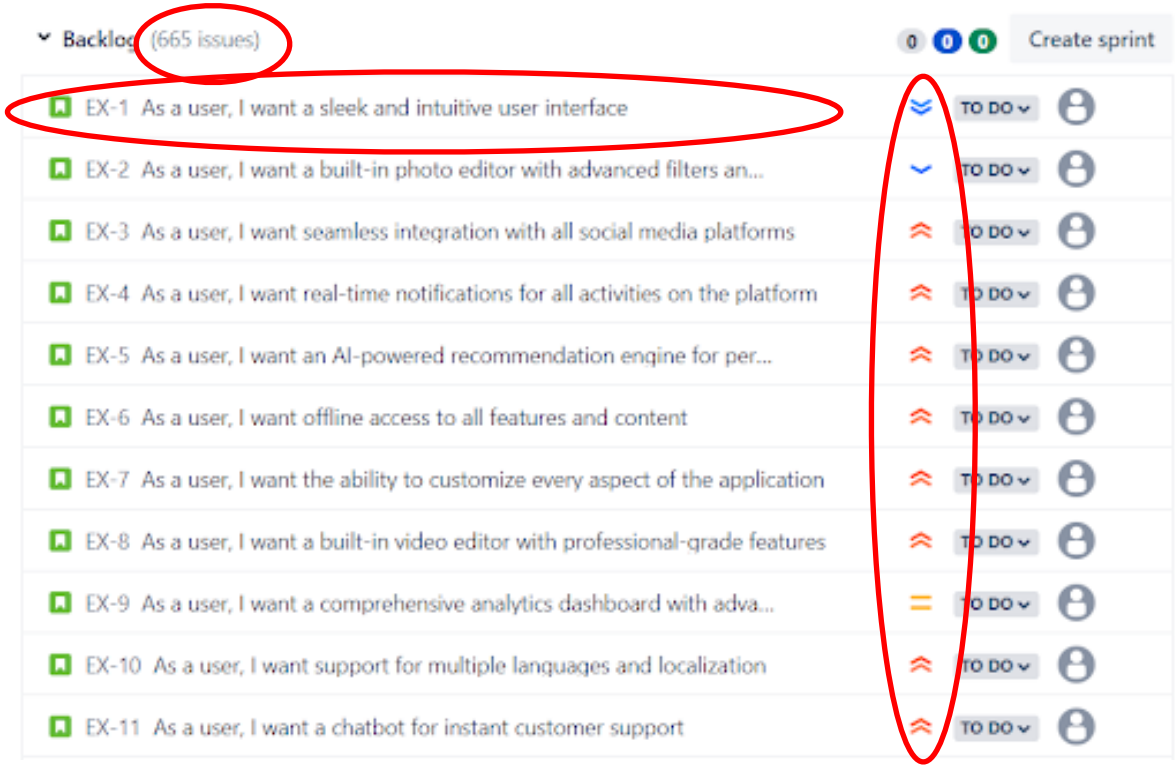
As a <type of user > I want <some feature, capability of the system> so that <I achieve some goal>.

A specific type of user with a specific, identifiable job

A capability, function or feature that is required for the system to support the user

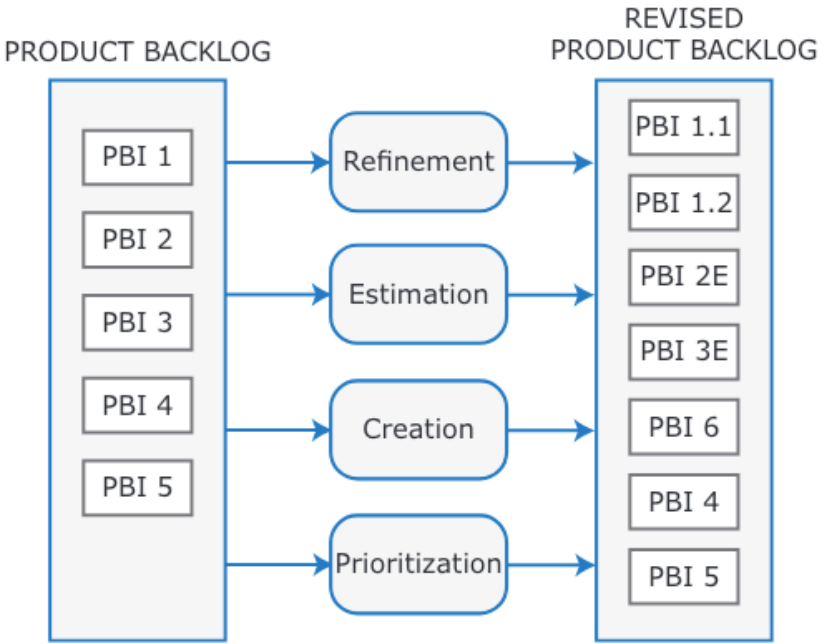
The goal the user is looking to achieve, or the value they are looking to derive using the system

Product Backlog

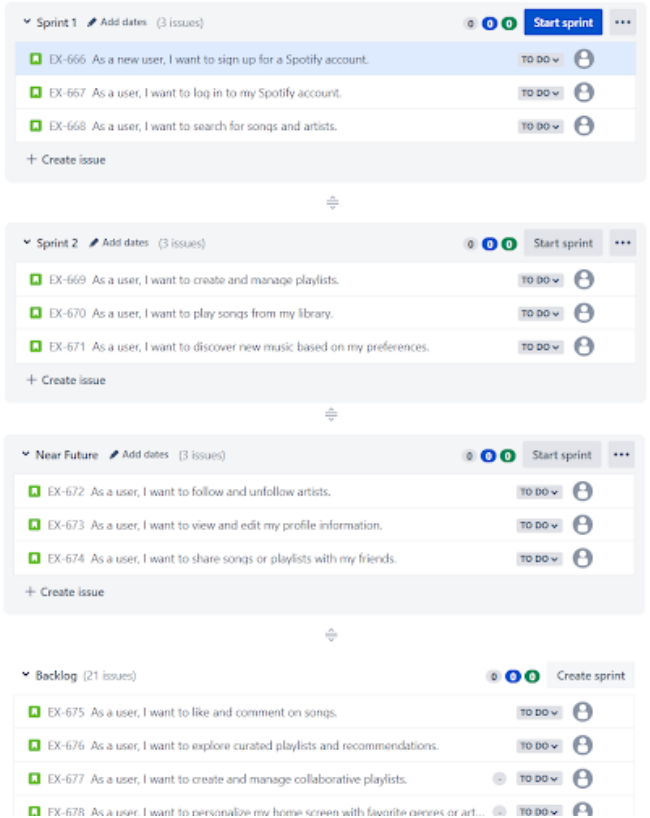


https://theproductmanager.com/topics/product-backlog-examples/

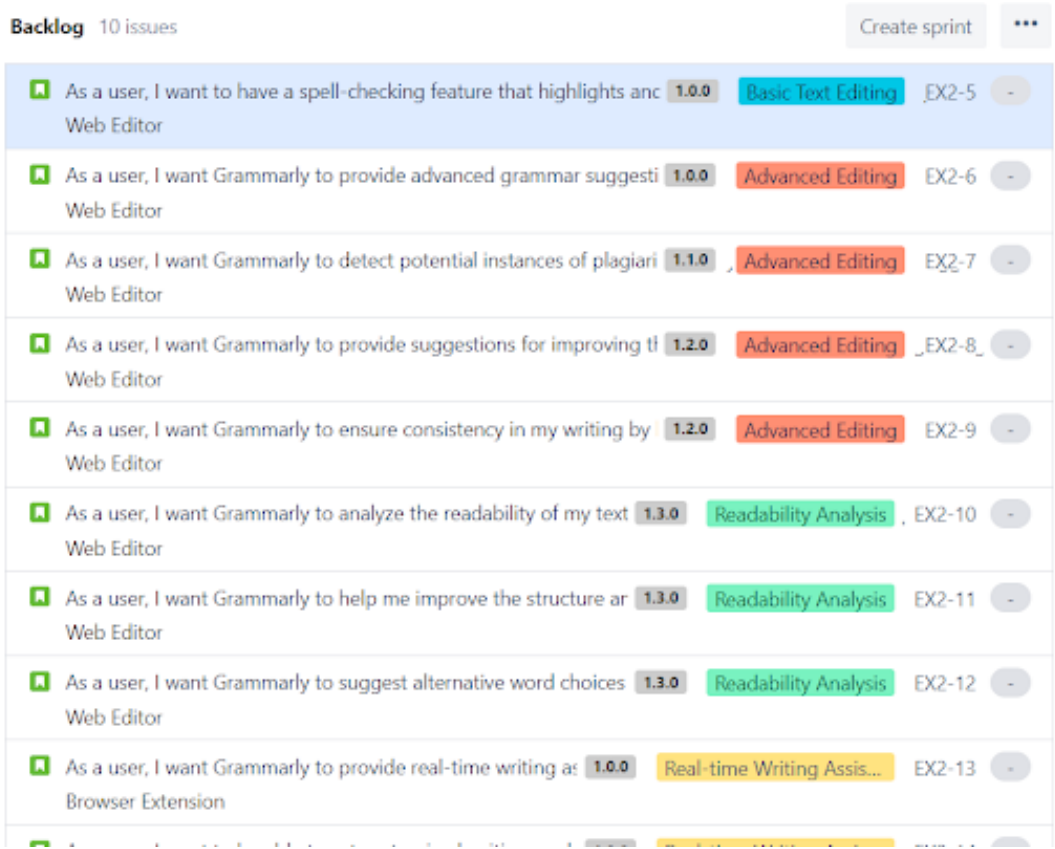
Product Backlog



Product Backlog



Product Backlog



Product Backlog

User Story: As a user, I want to sign up for a new account.

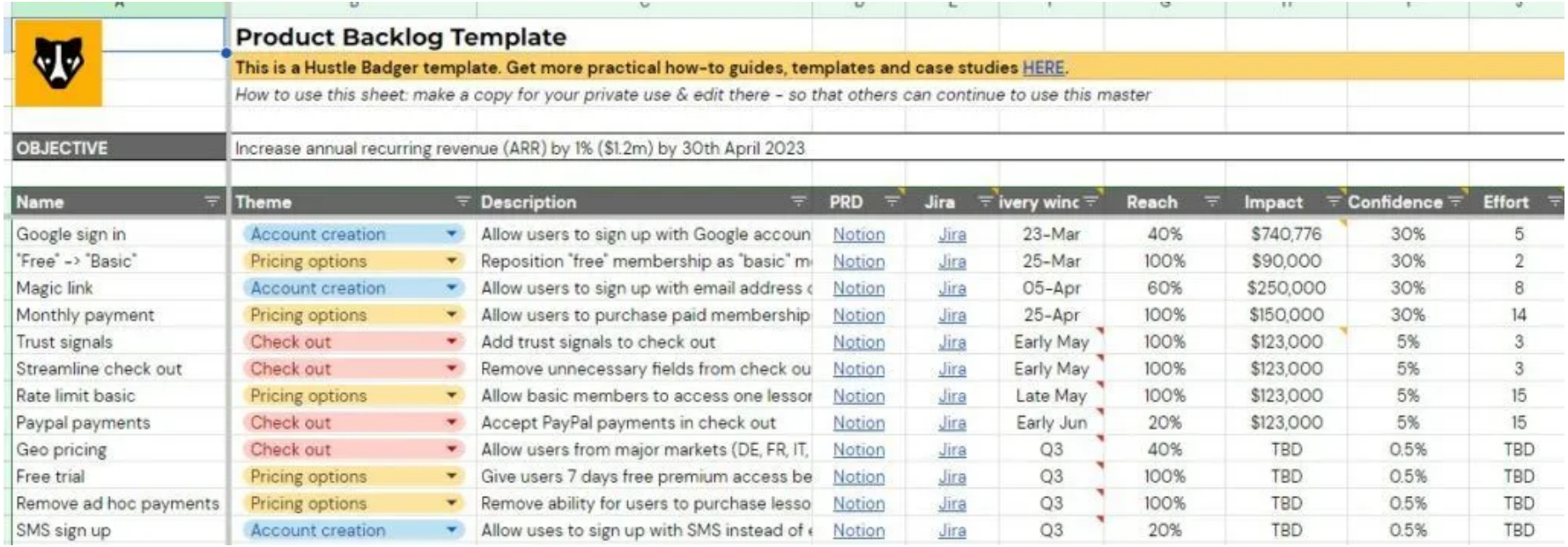
Requirements:

- Implement a registration form with the following fields:
  - Username: Allow the user to enter a unique username for their account.
  - Email: Provide a field for the user to input their email address.
  - Password: Include a password field for the user to set their account password.
- Implement password validation rules to ensure security:
  - Minimum password length of 8 characters.
  - Requirement for a combination of uppercase and lowercase letters, numbers, and special characters.
- Email Verification:
  - Integrate an email verification process for account activation.
  - Send an email to the provided email address with a verification link.
  - Upon clicking the verification link, verify the email and activate the user's account.
- Handle Error Cases:
  - Implement error handling for scenarios such as duplicate usernames or invalid email addresses.
  - Display appropriate error messages to the user and prompt them to correct the information.

User Story: As a user, I want to integrate my images with Social Media Platforms.

Requirements: We want the user to be able to directly post images to social media. We're currently looking at integrating with Instagram and Twitter.

Product Backlog



Product Backlog Template

This is a Hustle Badger template. Get more practical how-to guides, templates and case studies HERE.

How to use this sheet: make a copy for your private use & edit there - so that others can continue to use this master

OBJECTIVE	Increase annual recurring revenue (ARR) by 1% ($1.2m) by 30th April 2023

Name	Theme	Description	PRD	Jira	ivery winc	Reach	Impact	Confidence	Effort
Google sign in	Account creation	Allow users to sign up with Google accoun	Notion	Jira	23-Mar	40%	$740,776	30%	5
"Free" -> "Basic"	Pricing options	Reposition "free" membership as "basic" m	Notion	Jira	25-Mar	100%	$90,000	30%	2
Magic link	Account creation	Allow users to sign up with email address	Notion	Jira	05-Apr	60%	$250,000	30%	8
Monthly payment	Pricing options	Allow users to purchase paid membership	Notion	Jira	25-Apr	100%	$150,000	30%	14
Trust signals	Check out	Add trust signals to check out	Notion	Jira	Early May	100%	$123,000	5%	3
Streamline check out	Check out	Remove unnecessary fields from check ou	Notion	Jira	Early May	100%	$123,000	5%	3
Rate limit basic	Pricing options	Allow basic members to access one lesso	Notion	Jira	Late May	100%	$123,000	5%	15
Paypal payments	Check out	Accept PayPal payments in check out	Notion	Jira	Early Jun	20%	$123,000	5%	15
Geo pricing	Check out	Allow users from major markets (DE, FR, IT,	Notion	Jira	Q3	40%	TBD	0.5%	TBD
Free trial	Pricing options	Give users 7 days free premium access be	Notion	Jira	Q3	100%	TBD	0.5%	TBD
Remove ad hoc payments	Pricing options	Remove ability for users to purchase lesso	Notion	Jira	Q3	100%	TBD	0.5%	TBD
SMS sign up	Account creation	Allow uses to sign up with SMS instead of	Notion	Jira	Q3	20%	TBD	0.5%	TBD

https://www.hustlebadger.com/what-do-

Formulating a task

<*action*> **the** <*result*> <**by**|**for**|**of**|**to**> <*object*>

Product Backlog

Table 2.9 Code completeness checklist

State	Description
Reviewed	The code has been reviewed by another team member who has checked that it meets agreed coding standards, is understandable, includes appropriate comments, and has been refactored if necessary.
Unit tested	All unit tests have been run automatically and all tests have executed successfully.
Integrated	The code has been integrated with the project codebase and no integration errors have been reported.
Integration tested	All integration tests have been run automatically and all tests have been executed successfully.
Accepted	Acceptance tests have been run if appropriate and the Product Owner or the development team has confirmed that the product backlog item has been completed.


Kahoot!

Joining GitHub Classroom

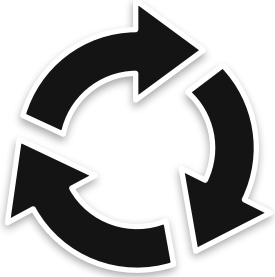
https://classroom.github.com/a/GqkA5iUB

https://forms.office.com/e/R7cs9MT57K

Managing technical risks



People

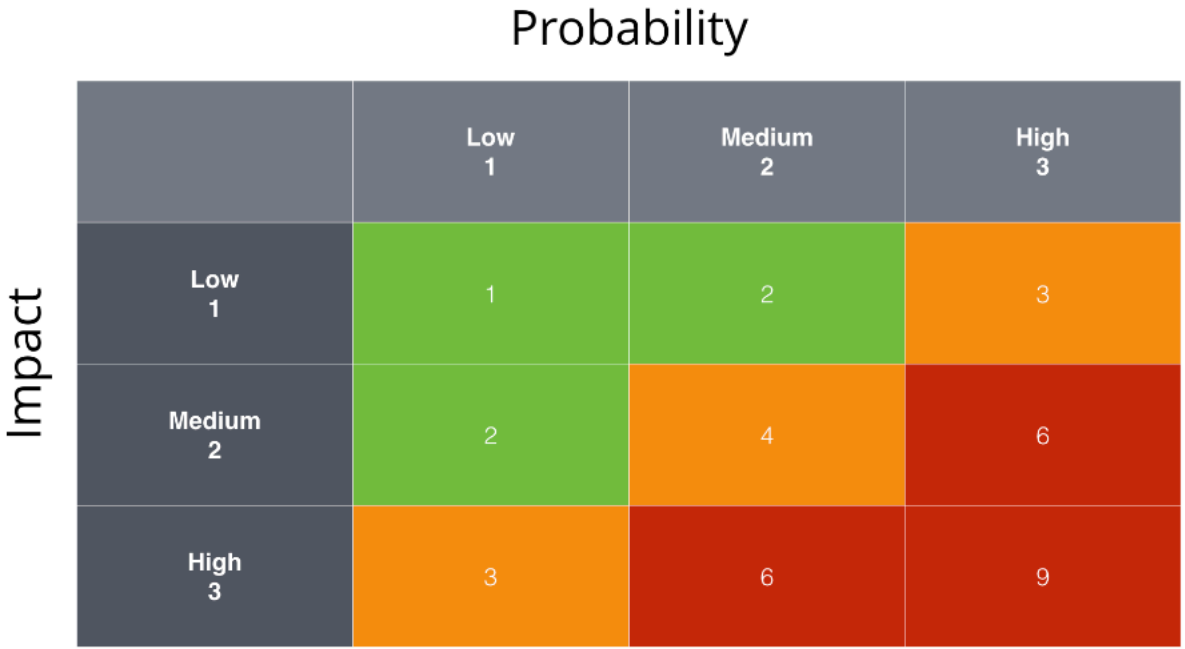


Process



Technology

Managing technical risks



A probability/impact matrix for quantifying risk

Actionable points

Identify main functional Software Requirements

Create User Personas

Build Features

Build some User Stories

Build the Backlog

Next time on …

Finish onboarding Form and GitHub Classroom

Course 3

Please read from *Engineering Software Products (Sommerville)*:

- Chapter 4 – Software Architecure (p.92-104)

Please read from Software Architecture for Developers (Brown):

- Chapter 4 - What is “software architecture”? (p.6-11)

